AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY (AUST)
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Department of Computer Science and Engineering
Program: Bachelor of Science in Computer Science and Engineering

# Project Final Report

Course No: CSE-1200
Course Title: Software Development – I

Project Title: Retro-Metro

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Submitted to

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# Introduction

## 1.1 Project Overview

Retro-Metro is a 2D combat to guide the player through multiple interconnected levels of a metro station, engaging in combat with security forces, earning scores, and overcoming environmental challenges, ultimately reaching the control room to solve a puzzle and shut down the metro system.

## 1.2 Objective

- To develop an interactive 2D game using the iGraphics library.
- To implement fighting mechanics, movement mechanics, and scoring systems.
- To create an engaging experience with increasing difficulty and boss battles.

# Game Development

## 2.1 Tools & Technologies Used

- **Programming Language:** C, C++
- **IDE:** Visual Studio 2013
- **Graphics Library:** iGraphics
- **Platform:** Windows PC

## 2.2 Core Features

- **Player Movement:** Controls the player" Kakashi" moving horizontally and jumping vertically.
- **Shooting Mechanics:** Players can shoot "Surikens" to destroy incoming aliens.
- **Wave System:** Difficulty increases with progressive waves.
- **Power-Ups:** Potions for health recovery. After defeating a boss, "Kakashi" will gain a new power for the next level.
- **Scoring System:** Players earn points based on enemies defeated.
- **Boss Battles: Final wave** features a powerful final boss of that level.
- **Environmental Obstacles:** There are few obstacles which will restrict player movement.

- **Levels:** Every level has unique enemy movement
- **Game Over Conditions:** The game ends if the health bar of “Kakshi” hits zero.

# Implementation Details

## 3.1 Code Structure

The game consists of multiple modules:

- **`main.cpp`** – Initializes the game, starts the iGraphics engine.
- **`loading.cpp-`** Handles the home screen animation.
- **`menu_Screen.`**cpp- Full menu implementation is handled in this portion of the code. It is tied with helper modules. Those are:
  - **`Chosse_screen.`**cpp
  - **`Credits_screen.`**cpp
  - **`Hep_screen.`**cpp
  - **`Settigns_screen.`**cpp

- **`level_1.cpp-`** The whole level one is handled by this module which includes player movement, boss mechanics, enemy movement.
- **`level_2.cpp-`** The whole level two is handled by this module which includes player movement with powerup, new boss mechanics, enhanced enemy movement.
- **`level_3.cpp-`** The whole level three is handled by this module which includes player movement with powerup, new boss mechanics, enhanced enemy movement.
- **`level_4.cpp-`** The whole level 4 is handled by this module which includes player movement with powerup, new boss mechanics, enhanced enemy movement.

## 3.2 Challenges Faced & Solutions

| Challenge | Solution |
|---|---|
| Player was moving beyond the background sprite | Restricted the movement within the size of the background sprite |
| Player movement was laggy. | Adjusted frame update rate for smooth movement. |
| Enemy spawning was predictable. | Implemented randomized enemy formations. |
| Game was taking a lot of time to start | Divided all the assets level by level. When level one starts only level one assets will be loaded. Not all assets will load at once. |

# Screenshots

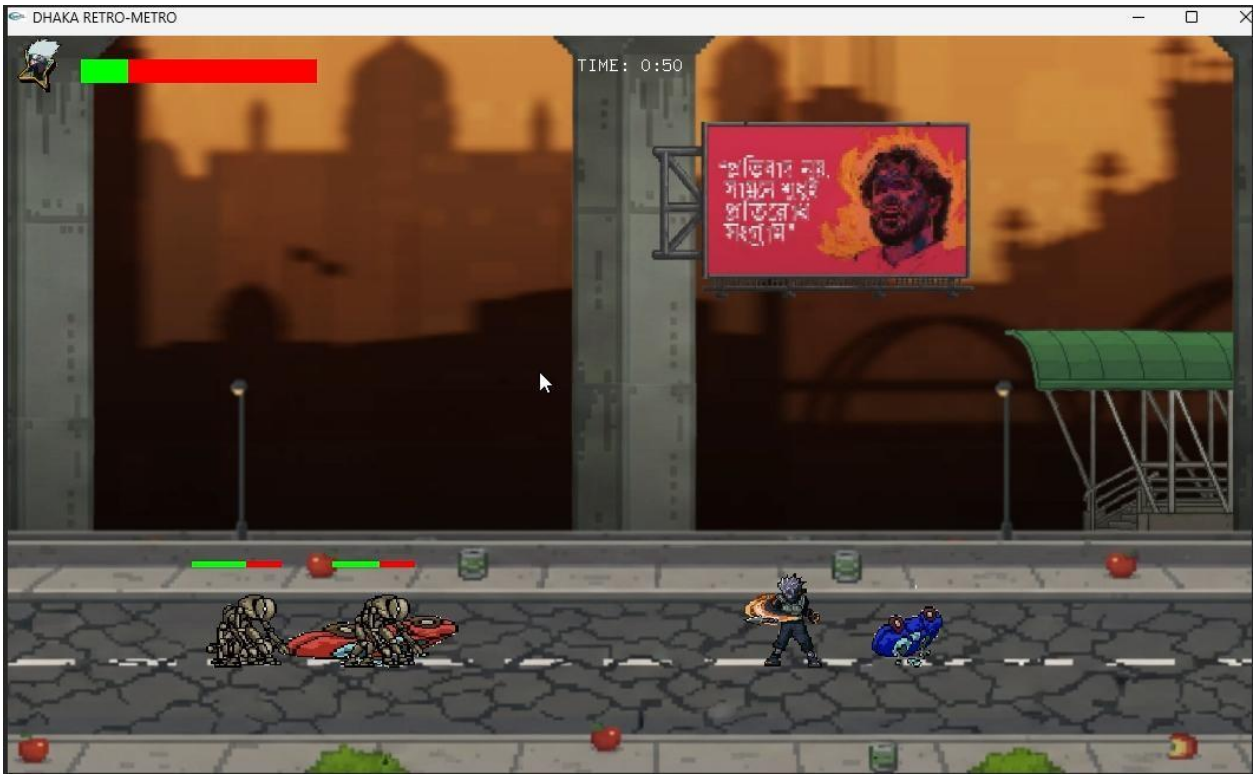



**Figure 1: Hero Attacking (Level One)**

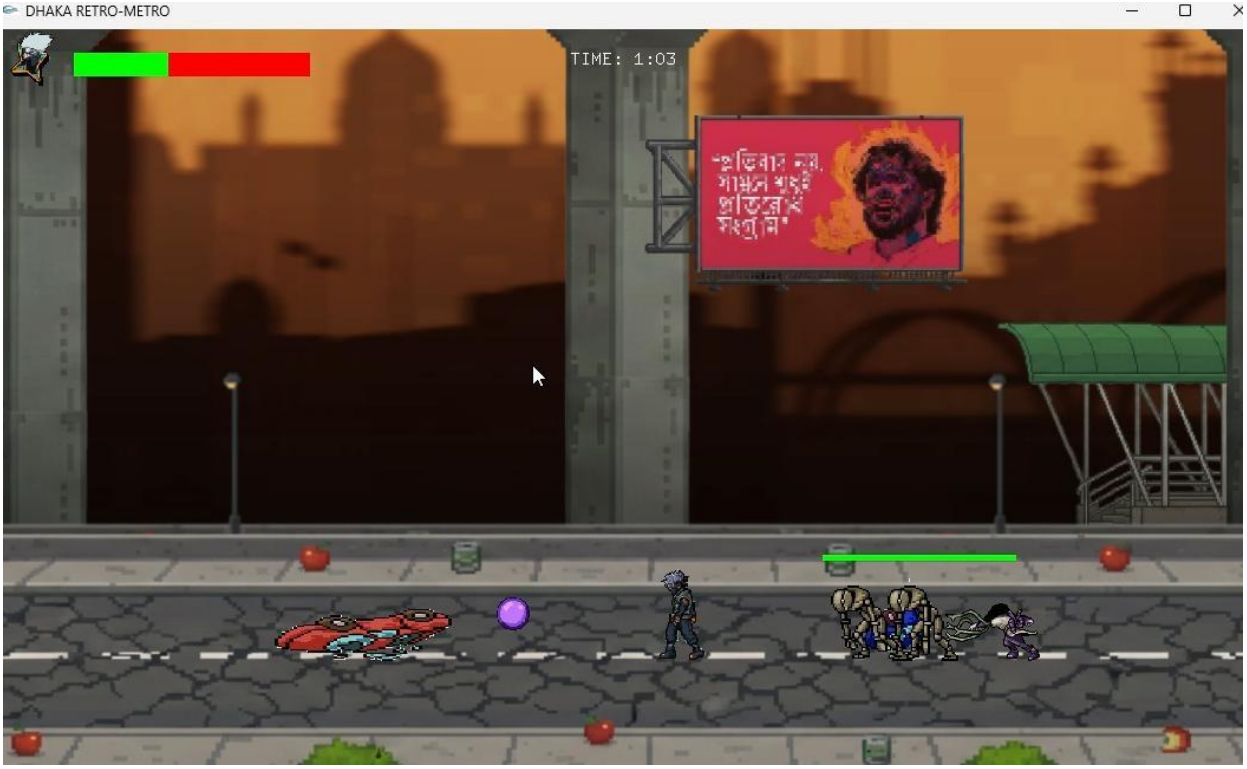



**Figure 2: Boss Fighting (Level One)**

**Figure 3: Close Combat (Level Two)**

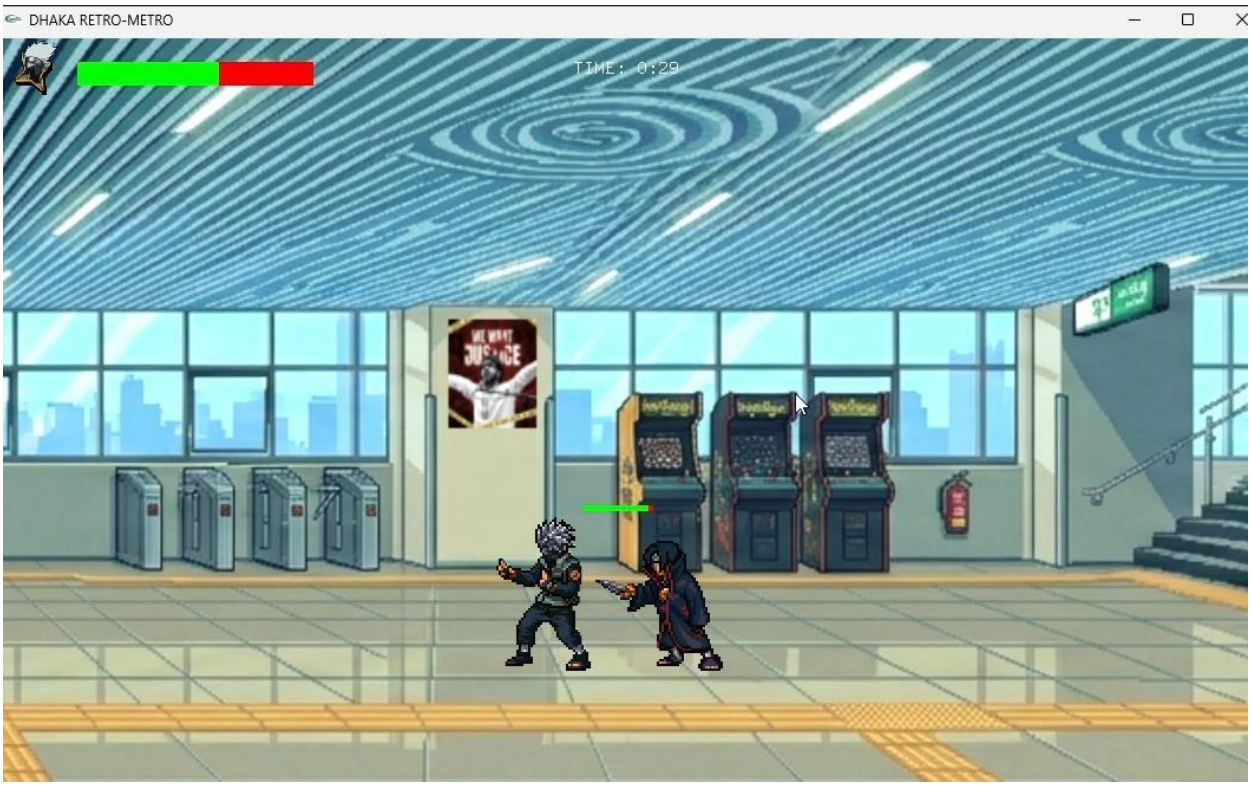



**Figure 4: New Boss Fight (Level Two)**

# 4. GitHub Collaboration

## 4.1 Repository Management

The project was managed using GitHub to enable efficient collaboration, version control, and code sharing among team members. A dedicated repository was created where all source code, assets, and documentation were maintained. However as all the members of the group weren't familiar with "GitHub" and git function and they contributed their part via **Google Drive**. Hence some irregularities in commits can be noticed.

Able team member contributed through:

- Creating and updating files via commits
- Working on separate modules using branches
- Merging changes using pull requests
- Resolving conflicts collaboratively

Those who were unable to contribute through git, contributed in the project by sharing their portion of work via **Google Drive.**

## 4.2 Contribution Workflow

The team followed a structured workflow:

1. Repository initialized with project structure
2. Members cloned the repository to their local systems
3. Features were developed in separate branches
4. Changes were committed with meaningful messages
5. Pull requests were created and reviewed
6. Approved changes were merged into the main branch

This workflow ensured:

- Proper tracking of contributions
- Reduced code conflicts
- Better team coordination

## 4.3 Individual Contributions

Each member was responsible for specific modules:

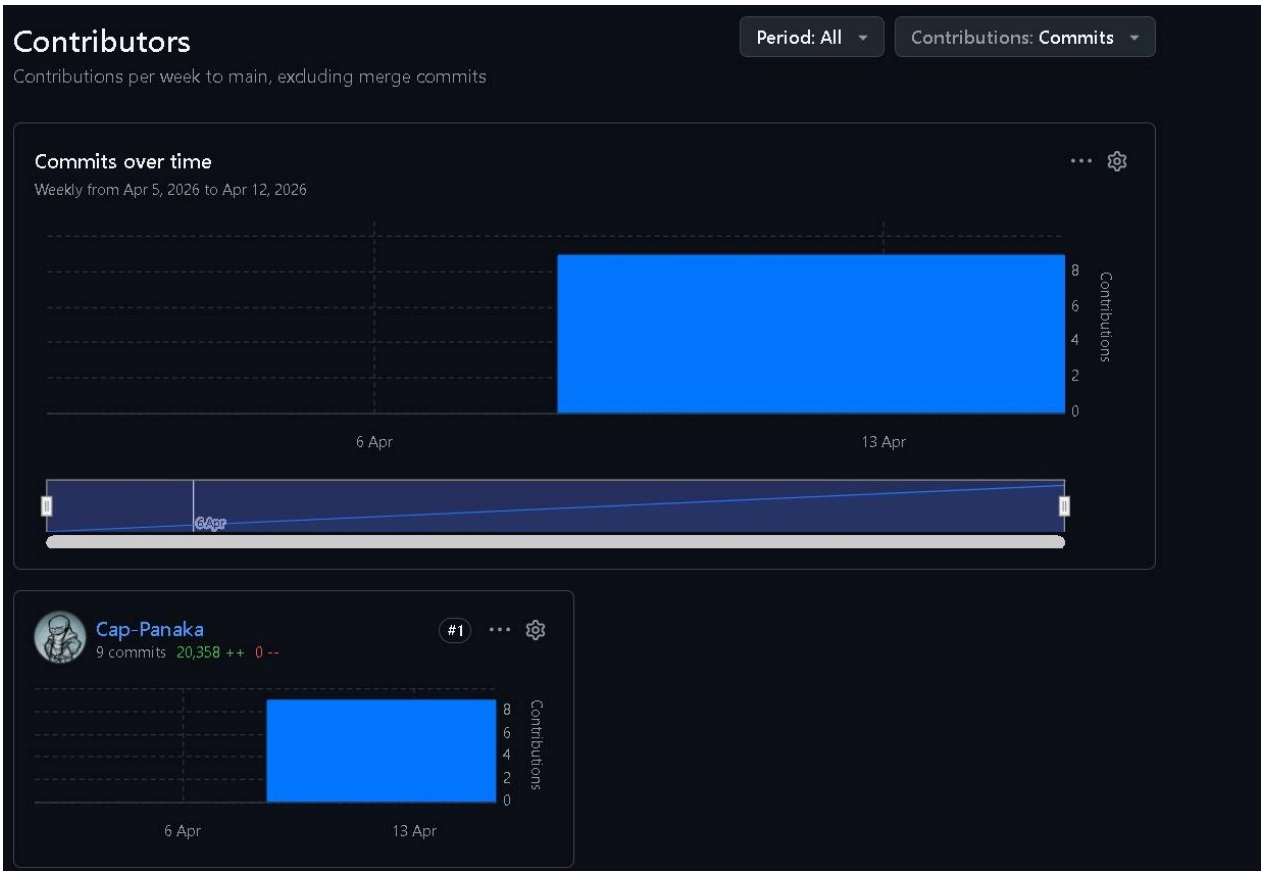
- **RIFAT AHSAN RIF (00724205101042):**
  Implantation of the boss enemy's movement for each level. Game assets collection and creation of different assets for the game. Designed the menu screen and it's hovering effect.

- **MD. FAZLE RABBI (00724205101046):**
  Implantation of player movement system and level transition. Implemented hidden code in each level which is required for the final level. Added a soduku in 3rd level.

- **MD. SADMAN ZUHAYER RAKIM (00724205101048):**
  Designed the home screen menu. Connected all options in the menu section with it's respective usage except "continue". Implantations of story panels between levels. Implantation of UI of health bar and mana bar and time count

- **MD. NAFIZ KHAN(00724205101064):**
  **Implantation of players attack system and special attack system. Implemented potions for increasing health after defeating a certain amount of enemy. Implemented the "continue" function in the menu bar**

## 4.4 Contribution Evidence (Screenshot Required)




**Cap-Panaka** is the Git-Hub account of **Nafiz Khan(00724205101048)**

**RIFAT AHSAN RIF(00724205101042), MD. FAZLE RABBI(00724205101046**) and **MD. SADMAN ZUHAYER RAKIM(00724205101048)** their works via **Google Drive** which was then commited by Nafiz. Thus this irregularity in the photo.

# Conclusion & Future Improvements

## 5.1 Summary

Retro-Metro project successfully implemented a 2D arcade-style game with smooth mechanics and interactive gameplay. iGraphics allowed for real-time rendering of graphics and animations.

## 5.2 Future Enhancements

- **Multiplayer Mode** – Allow two players to compete or cooperate.
- **Sound Effects** – Add background music for each level and in the boss battle.
- **New Characters**- Add new main characters for better immersion.
- **Advanced Graphics** – Improve enemy and main character animations and backgrounds.